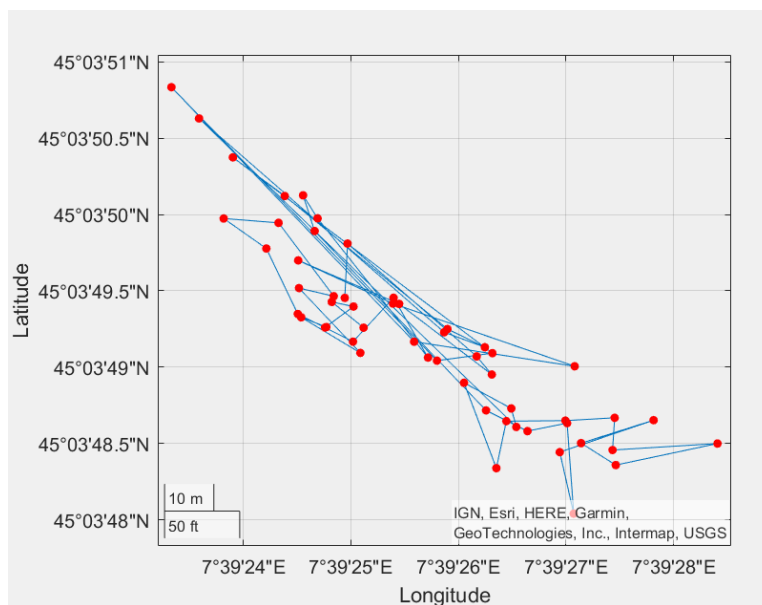


GNSS Lab Report Guidelines

1. General Rules and approach

- Abide by the rules outlined in the course intro slides
- Keep it short: **no more than 3 pages** (references and appendix can be on the 4th page)
- Select only the **meaningful plots** that are relevant with respect to your comments (focus on your own data collection)
 - You are strongly encouraged to modify the plots (e.g., add superpositions, zoom-ins, etc.) to highlight key insights and improve conciseness.
- In the report it's crucial to provide a clear and concise overview of the experiment's **purpose, methods, results, and conclusions**. However, we are not interested in a mere repetition of the theoretical concepts from classes, rather we are interested into an insightful analysis of the results.
- If you make qualitative assessments, they **must** be substantiated.
 - For instance, avoid commenting a plot like the one in the figure claiming that the estimate is «not accurate». Rather, measure its dispersion (e.g. variance, standard deviation, root mean square) and/or compare it with other data collections



2. Report Structure and content

Your lab report should follow a scientific paper like structure and be written in a clear, concise, and technical style. The report should demonstrate your understanding of GNSS signal collection and the effects of spoofing (without repeating the theoretical lecture), as well as your ability to analyze and interpret data.

You should organize your report according to the following outline:

3 pages max!

1. **Methods**
2. **Results and Discussion**
 1. **Baseline data collection + spoofing applied**
 2. **[Optional] Peculiar conditions data collection**
3. **Conclusion**
4. **References**
5. **Appendix**

2.1 Methods (brief and schematic)

- Devices and tools used: List and describe the Android devices, apps (e.g., GNSS Logger), spoofing tools, and any post-processing code.
- Data collection procedure:
 - How and where the GNSS data was collected (e.g., open sky, indoors, under spoofing conditions).
 - Settings used during data logging (e.g., battery saving enabled, logging duration).
- Spoofing setup: Describe how spoofing was introduced and its settings/characteristics

2.2 Results and Discussion

- Data presentation:
 - Include plots, tables, and summaries of key results.
 - Follow lab tasks and **answer task questions**
 - Report outputs (plots, tables, etc) when relevant for the discussion.
- Interpretation:

- Discuss what the data shows about the quality of GNSS measurements and PVT under different conditions.
- When spoofing is performed, explain its effect on position estimates or signal metrics.
- Mention any anomalies, unexpected results, or challenges encountered.
- Limitations:
 - Discuss any sources of error or limitations in your setup or data interpretation.

2.3 Conclusion

- Summary & insights: Concisely recap the main findings and reflect on what the experiments reveal about GNSS and its vulnerabilities.
- Future work: Briefly suggest potential improvements or extensions for this kind of experiment.

2.4 General Formatting notes

- Include figures and tables with appropriate captions and references in the text.
- Cite any tools, apps, or relevant literature you used or referred to.
- Use technical terminology appropriately but define any concepts that are not commonly known (or reference proper sources)

6. Formatting Instructions

Prepare a report using the LaTeX conference-style format for ACM CCS that you can find on www.overleaf.com.

- Overleaf is a free online editing system for LaTeX. It allows collaborative editing, with a simple interface and visual editor based on LaTeX, the most used markup system to describe scientific documents.
- LaTeX is a markup system in the early 1980s by Leslie Lamport who extended the original TeX first released in 1978 by Donald Knuth. Since then, it has been the de facto standard for producing scientific documents. It lets you focus on the content and not on the pagination. There are plenty of LaTeX tutorials, e.g., [the Overleaf help pages](#).

You need to create a free account on Overleaf, then create a new document by looking for the Sample ACM CCS template.

Next, modify the document format to

- use the “review” option: `\documentclass[sigconf, review]{acmart}`
- change authors, title, etc.
- remove the copyright part to save space:
 - `\setcopyright{none}` %remove the (c) section
 - `\acmConference{Wireless Security Report}{Torino}{2024}` % set a possibly significant conference name
 - `\acmPrice{}` % leave the rest empty. These will still appear.
 - `\acmISBN{}`
 - `\acmDOI{}`

You can share the document using the [Share with link](#) option so that anyone with the link can collaborate to edit the file.